

EEG x Deep Learning: what is in the signal, and what the networks learn from it

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Every number below is produced by `python main.py` (seeds fixed, every run reported in the CSVs named in brackets). Rationale for each choice is in `DECISIONS.md` (D-numbers), and predictions with their scorecards in `PREDICTIONS.md`.

Task 1: Motor imagery, subjects 70-109

A1 Data audit

Runs 4, 8 and 12 are imagined left vs right fist (T1 = left, T2 = right). Runs 3/7/11 are execution and 6/10/14 imagine fists vs feet, so the event codes mean different things per run (D1, verified in the `mne.datasets.eegbci` docstring and on the PhysioNet page). `audit.csv` has one row per subject. Three subjects (88, 92, 100) carry a 128 Hz header. They were kept and resampled to 160 Hz because their alpha peak does not show the 0.8x shift a mislabelled 160 Hz file would give, and their protocol differs consistently (19 trials per run, 5.12 s cues), so the header is right (D2). Twenty trials whose -1.5 to 4.0 s window runs past the end of the recording are dropped and listed per subject (D3), leaving 897 left and 896 right trials.

A2 Physiological ground truth

The recording reference is undocumented. Every channel then carries the same unknown reference signal, and a lateral reference (e.g. a mastoid) would bias C3 and C4 unequally, so a raw C3-vs-C4 difference partly measures the reference. A local Laplacian (C3 or C4 minus the mean of its 4 nearest electrodes, D4, D10) has weights summing to zero, so any component common to the neighbourhood (the reference included) cancels, and the comparison becomes local. Power is Morlet time-frequency power at C3, C4 and their neighbours, relative to a pooled pre-cue baseline (-1.0 to -0.1 s, D5), averaged over 0.5-4.0 s. The lateralisation index $LI = (|ipsi| - |contra|) / (|ipsi| + |contra|)$ per band, tested with a one-sided label-permutation test (10 000 permutations, Bonferroni over mu and beta) and labelled present only if the contralateral change is a desynchronisation (D6). **14 of 40 subjects show lateralised ERD** (`a2_lateralisation.csv`; mean LI 0.26 mu, 0.32 beta). CAR and raw referencing give smaller LIs (0.19/0.29 and 0.15/0.21, `a2_all_references.csv`).

A3 Confound audit

The cue is a target on one side of the screen, shown until the subject stops imagining, with rest between trials. Each non-motor route got its own restricted decoder (D12-D17): eye movement (frontal channels, 0-0.5 s and 0.5-4 s), visual response (occipital), muscle (temporal 30-40 Hz), trial order and history (no EEG), and pre-cue twins of every EEG decoder as a leak check. Threshold: exact one-sided binomial at 5% (63-67% within subject, 52.0% pooled over 1793 trials). A permutation null gave 2.7-8.3% false positives against the nominal 5% (`outputs/A3_RESULTS.md`).

decoder	within-subject mean	subjects above threshold (of 40)	pooled cross-subject	pre-cue twin
F1 frontal, 0-0.5 s (eye)	66.4%	25	71.9%	48.2%
F2 frontal, 0.5-4 s	63.5%	18	75.3%	-
O1 occipital, 0-0.5 s (visual)	56.5%	14	59.3%	49.7%
G1 temporal 30-40 Hz (muscle)	50.8%	1	49.6%	49.3%

decoder	within-subject mean	subjects above threshold (of 40)	pooled cross-subject	pre-cue twin
M1 sensorimotor mu+beta (motor)	54.9%	8	54.6%	52.2%*
R2 previous 1-3 labels (no EEG)	70.8%	31	72.6%	-

*Marginal (permutation p 0.02-0.03 among 10 uncorrected pre-cue twins), not interpreted.

The prediction (eyes and visual decodable but weak, motor 56-62%, order at chance) was wrong on size and on order: the eye route dominates, labels alternate ($z = -17.6$), and the zero-phase filter first made the pre-cue window look decodable until a causal filter removed the leak (D18). The per-subject table with LI, p, label and every decoder accuracy with its threshold is `a3_per_subject_table.csv`.

B1 Models and evaluation

Cross-subject only: 8 fixed subject-wise folds of 5, test = fold f, validation (early stopping only) = fold f+1, normalisation from training subjects only (D22). B2 uses folds 0-3 as test (20 subjects, 913 trials, threshold 52.8%), seeds 0-2 everywhere (D24). EEGNet-8,2 (2962 parameters) is the 128 Hz design with both 0.5 s kernels rescaled to 160 Hz (64 to 80 and 16 to 20 samples), so the lowest shapeable frequency stays at 2 Hz (D20). The transformer (d_model 16, 2 layers, 2 heads) shares EEGNet's 0.5 s temporal filter and changes only the token axis: time-patch tokens (6834 parameters), channel tokens with (7250) or without (6226) a learned electrode identity, all within 3x EEGNet (D21).

model	5 subj.	10	20	30	30, eye route removed
EEGNet	72.1 +- 0.5	75.0 +- 1.3	76.2 +- 1.3	76.9 +- 1.0	50.5 +- 2.9
Transformer, time-patch	54.8 +- 3.1	64.3 +- 3.0	66.8 +- 4.1	71.6 +- 3.2	49.1 +- 1.3
Transformer, channel + identity				70.2 +- 1.2	
Transformer, channel, no identity				57.5 +- 1.6	

Test accuracy %, mean +- SD over 3 seeds. Every fold x seed run is in `outputs/partb/b2_all_runs.csv`. Displacement, attribution, faithfulness and the intervention are in `outputs/partb/B2_RESULTS.md` and figures `b2_*.png`.

Write-up

(a) If a model scores 70% on these subjects, how much of that needs motor imagery?

Almost none. The cue is a target on the left or right of the screen, and the subject looks at it. A decoder restricted to frontal electrodes in the first 0.5 s after the cue (the saccade window, before ERD develops) reaches 71.9% across subjects and is above threshold in 25 of 40 subjects. Its pre-cue twin is at chance (48.2%), so the information arrives with the cue and is not a filter artefact. In contrast, the sensorimotor strip (mu and beta, 0.5-4 s) reaches 54.6% pooled and 8/40 subjects, although A2 finds lateralised ERD in 14/40. The motor signal is real but inconsistent across subjects, while the eye signal has the same sign in everyone.

The networks follow this. EEGNet reaches 76.9% and the time-patch transformer 71.6% on all 64 channels. With the eye route removed (local Laplacian, then the 21 Fc/C/Cp electrodes; D25) both fall to chance (50.5%, 49.1%). The problem is not that they cannot fit motor imagery, it is that what they fit does not transfer: training loss falls while validation-subject loss rises from the first epoch (D27). Attribution agrees: per electrode, both rely most on F7, F8, Ft7, Ft8 and Af8 (horizontal EOG positions), and the transformer peaks at 0.2-0.4 s.

A second route sits in the protocol. Consecutive labels alternate (424 repeats against 780 expected, $z = -17.6$), and the previous label alone predicts the current one at 72.6% with no EEG. A single-trial network cannot use it directly (the previous label is not readable from the EEG features, 46-51%), but any evaluation that leaks neighbouring-trial context inherits it.

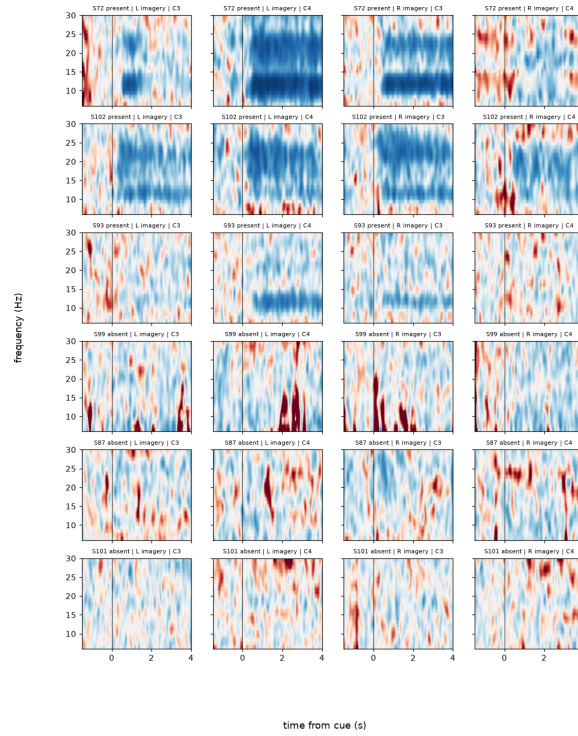


Figure 1: Laplacian-referenced time-frequency power relative to baseline at C3 and C4, for left and right imagery (blue = desynchronisation). Top three rows: lateralised ERD present (S72, S102, S93). Bottom three: absent (S99, S87, S101).

So 70% is fully explainable without motor imagery: the eye route alone gives 72%, the motor route under 55%. The exact size depends on the pre-filter (a causal Butterworth gives 65.3% for the frontal decoder, since part of the eye signal is below 1 Hz), but the conclusion survives both filters. A motor-imagery claim on this dataset should report accuracy after the eye route is removed, and here that number is chance.

(b) Mechanisms behind the B2 results

Scaling (predicted crossover: wrong). EEGNet leads at every size (72.1% at 5 subjects, 76.9% at 30) and the transformer closes the gap from ~17 to ~5 points without crossing. EEGNet hard-codes the problem (a shared temporal filter bank, one spatial filter per band, pooled power), so ~2900 weights in that fixed form pin down the frontal eye pattern from five subjects. The transformer shares the front end but must learn how 20 time segments combine, and that freedom costs data. A crossover needs a signal richer than EEGNet’s form can express, and a spatially broad eye field is not one.

Tokenisation (predicted time > chan+id = chan-noid: partly wrong). Time-patch (71.6%) and channel tokens with identity (70.2%) are within seed spread, but dropping the identity embedding costs ~13 points (57.5%). Without identity, attention plus a CLS read-out is permutation-invariant over electrodes, so it cannot represent where a signal is: ERD under C3 and under C4 are the same input set. Tested directly, its predictions are 100% unchanged under electrode shuffling and left/right mirroring. This matters because the label is itself a mirror variable (contralateral ERD and saccade direction both flip with side), so its 57.5% can only come from mirror-asymmetric parts of the recording, such as an off-centre reference.

Displacement (predicted time-patch drops most: right in ranking). A 10 mm slide (a quarter of the electrode spacing) costs every model under one point, 20 mm at most 1.5 points. EEGNet’s spatial filters are sensitive in isolation ($|W(M-I)|/|W| = 0.41$ at 10 mm), but the projected signal is smooth over 10 mm. Layer by layer, EEGNet’s perturbation grows in the depthwise spatial layer (13% to 17%) and temporal pooling averages it back to 13% at the logits, while the time-patch transformer’s grows at every stage to 17%, because its spatial filters feed every token and the softmax amplifies a small coherent change.

Faithfulness (predicted attention > gradient > random: right). Deleting the most-attended half of the tokens drops accuracy from 71.4% to 57.7%, against 64.1% for gradient x input and 65.6% for random. Gradient x input is a first-order estimate at the intact input, but deleting a token renormalises the softmax, which last-layer CLS attention captures directly.

Link to Part A (predicted LI-driven, small drop: half right). Per test subject, EEGNet accuracy correlates with the A2 lateralisation index (Spearman 0.72) and the A3 frontal decoder (0.53), which are unrelated to each other (-0.02). The network uses both

routes, in different subjects. However, removing the eye route drops it to chance, not by a few points. This is the largest contradiction, and its cause is transfer, not presence: the motor pattern is subject-specific in strength and topography, and the eye pattern is the only one shared.

Intervention (predicted no effect: right). Training with random cap slides (0-15 mm) adds, to first order, the penalty ($\sigma^2/2$) $E[(\text{grad}_x L \cdot J_x)^2]$, the loss change along a slide. It is only large if the loss depends on exact electrode positions, which the displacement experiment rules out. Accuracy (77.7% vs 76.9%), the displacement curve and filter sensitivity (0.40 vs 0.41) are unchanged.

In short, a cross-subject network on this dataset learns motor imagery only if the shared non-motor route is removed and the motor pattern is consistent enough across subjects to transfer, and on these 40 subjects the second condition fails.

Task 2: Sleep staging pipeline

2a Baseline

Given script, unchanged, seed 42 (task2/baseline/2a_baseline_seed42.txt): **accuracy 0.847, macro-F1 0.628, kappa 0.693** (per-class F1 W 0.945, N1 0.207, N2 0.703, N3 0.642, REM 0.641). 70% of its test windows are Wake.

2b Defects

Ten defects, one commit each; what, evidence, distortion and fix are in DEFECTS.md, evidence printouts in task2/evidence/, judgement calls in DECISIONS.md (T2-1 to T2-10).

#	defect	evidence (not only code reading)	fix
1	split by recording	5/5 test subjects' other night in training (seed 42)	split by subject, 9/3/3
2	epoch chosen on test	test F1 0.58-0.67 over 12 epochs, max reported	choose on validation subjects
3	batch_first=False	own neighbour moves output by 0, another window by 9e-4	batch_first=True
4	position added after attention	shuffling neighbours changes output by 3e-7	encode before attention
5	label = last epoch, head = centre	labels differ in 8.7% of windows, 71% of N1	label the centre epoch
6	500 "uV" vs data in volts	0 of 80070 epochs rejected; units-only fix rejects 54% (EOG)	uV, EEG channels only
7	rejected epochs deleted	windows span time gaps	mask, skip such windows
8	z-score per epoch	99.6% of N3 epochs >75 uV; after scaling all std = 1	z-score per recording
9	unknown labels -> Wake	905 "?" + 35 movement epochs became Wake	label -1, context only
10	~23 h recordings	68% of epochs lights-on Wake; "always Wake" = 68.7%	crop to sleep $\pm$ 30 min

Fixed pipeline, mean $\pm$ SD over seeds 42-44: **accuracy 0.827 $\pm$ 0.012, macro-F1 0.788 $\pm$ 0.015, kappa 0.769 $\pm$ 0.020.**

2c Impact ledger

Each defect re-introduced alone into the fixed script by reversing its edit (task2/ledger.py), seeds 42-44 (a seed also draws the subject split); Δ = defective - fixed on the same seed, mean $\pm$ SD (task2/ledger_table.md, per run task2/ledger_runs.csv). Predictions were committed before any run (PREDICTIONS.md). Every single defect has 3 seeds;

the four combination rows marked † have seeds 42 and 43 only (their seed-44 runs were cut for time; the ALL run ran out of memory).

defect	Δ accuracy	Δ macro-F1	Δ kappa	predicted (F1)	outcome
10 no crop	+0.098 ± 0.017	+0.013 ± 0.018	+0.089 ± 0.028	acc up, F1 ~0	right
3 batch attention	-0.026 ± 0.022	-0.034 ± 0.020	-0.035 ± 0.027	down	right (N1 -0.089)
8 per-epoch z	-0.012 ± 0.010	-0.025 ± 0.019	-0.020 ± 0.013	down, N3 most	down, but N1 most
9 unscored -> W	-0.015 ± 0.007	-0.017 ± 0.007	-0.020 ± 0.006	~0	larger than predicted
2 select on test	+0.010 ± 0.009	+0.016 ± 0.014	+0.015 ± 0.013	up	right
1 recording split	-0.007 ± 0.023	-0.012 ± 0.008	-0.014 ± 0.036	up	wrong sign
4 position after	-0.003 ± 0.021	-0.006 ± 0.023	-0.003 ± 0.027	small down	within noise
5 last-epoch label	+0.009 ± 0.011	+0.007 ± 0.015	+0.013 ± 0.014	down	within noise
6 units (no rejection)	+0.006 ± 0.010	+0.008 ± 0.014	+0.009 ± 0.016	~0	right (noise floor)
7 deletion	+0.011 ± 0.019	+0.008 ± 0.015	+0.014 ± 0.024	~0	right (noise floor)
6N naive units fix †	-0.028 ± 0.004	-0.049 ± 0.002	-0.069 ± 0.000	down	right
6N + 7 †	-0.016 ± 0.015	-0.021 ± 0.026	-0.029 ± 0.024	worse than 6N	flips: less bad
3 + 4	-0.037 ± 0.016	-0.046 ± 0.031	-0.048 ± 0.024	= 3	right (4 vanishes)
1 + 2	+0.015 ± 0.033	+0.012 ± 0.026	+0.014 ± 0.053	more than either	not additive
9 + 10 †	+0.097 ± 0.012	+0.005 ± 0.017	+0.081 ± 0.021	9 grows	9 vanishes
all ten †	+0.027 ± 0.031	-0.147 ± 0.032	-0.059 ± 0.054	acc up, F1 -0.08..-0.20	right

Noise floor. Defects 6 and 7 change no test window here, and on seeds 42 and 43 their runs equal the fixed run exactly; on seed 44 both differ from it by the same +0.024 macro-F1. The difference is run-to-run nondeterminism (most likely the CPU thread count, which changes floating-point summation order), so effects under about ± 0.02 macro-F1 are not resolved.

Ranked by how much each would have misled the report. (1) **10**: it alone adds ~10 points of accuracy and 9 of kappa from lights-on Wake, while N1 gets worse; the headline would have been inflated by the easiest class. (2) **3**: the context model the architecture advertises never ran; the CNN alone still scores well, so nothing looks wrong. (3) **8**: removes the amplitude criterion of N3 and N1. (4) **9** and (5) **2**: consistent across all seeds but small. Defects 1, 4, 5, 6 and 7 are inside the noise floor on their own.

Flags. Defect 4 vanishes while 3 is present ($3+4 \approx 3$). Defect 7 vanishes while 6 is present (nothing is rejected, so nothing is deleted), and with the naive units fix it *flips*: deleting keeps 17.3k training windows where masking keeps only 11k (an 11-epoch window needs 11 kept epochs; Wake training windows fall from 2694 to 381, as waking eye movements trip the EOG threshold), so the original deletion hurts less than the “correct” masking. Defect 9’s cost on accuracy vanishes once Wake is uncropped (10), the opposite of the prediction: the mislabelled epochs drown in the extra Wake. Defect 1 went the wrong way: subject leakage did not inflate the score here, because the same seed also draws a different, harder set of test subjects, and the test set changes with it (D1 test windows 4750-5143 vs 5415-6653); with 3 test subjects, who is tested matters more than whether their other night was seen.

2d Physiology: the worst stage

Full analysis in `task2/PHYSIOLOGY.md`. The files hold EEG Fpz-Cz and Pz-Oz and horizontal EOG at 100 Hz; chin EMG, respiration and temperature only as 1 Hz envelopes; labels are R&K stages (3+4 merged into N3). The pipeline uses the EEG and EOG, not the EMG.

Worst stage: N1 (F1 0.511 ± 0.017 , others 0.83-0.89). Recall 0.61 but precision 0.44: the model uses N1 as the place for epochs it cannot place (over 3 seeds, 486 N2, 276 REM and 262 W windows called N1, against 832 true N1). N1 is defined by transitions (alpha falling below half the epoch, low-amplitude 2-7 Hz activity, slow eye movements, ended by the first spindle or K-complex), and on these channels it overlaps its neighbours: median Fpz-Cz amplitude 78 μV (REM 79), Pz-Oz alpha 0.084 (REM 0.088), slow-EOG power 284 μV^2 (REM 271). What tells N1 from REM is muscle tone (EMG 2.68 vs 0.69 μV), which the model never sees; what tells it from N2 is a single transient that a 30-s epoch dilutes. It is also the rarest stage (7% of scored epochs) and the one human scorers agree on least.

Expected confusion: N3 -> N2 (17.5% of N3). Stage 3 starts at 20% of the epoch with >75 uV, 0.5-2 Hz waves: a threshold on a continuum (median N3 coverage 27%), so borderline epochs fall to N2; the reverse is rare (3.2%). **Unexpected confusion: W <-> REM** (5.1% and 3.3%, 304 windows). Human scorers rarely confuse them, because REM requires chin atonia; the file records it (EMG 3.23 vs 0.69 uV, the largest separation of any feature), but the pipeline drops the channel, and on EEG + EOG REM looks like relaxed Wake with eye movements.

2e Note for a non-technical reader

The first version of our sleep-staging program reported 85% accuracy; the corrected version scores about 83%. The lower figure is the honest one.

The first test was too easy. Each recording runs for almost a whole day, and two-thirds of it is the person awake with the lights on. A program that always answered “awake” would already score 69%. The corrected test keeps only the night, plus half an hour either side.

The first test also included people the program had already studied (their other night was in its training material), and it picked its best version by peeking at the test. Real patients will be new to it.

On a balanced score that counts every sleep stage equally, rather than rewarding easy wakefulness, the corrected program improved from 0.63 to 0.79 (out of 1). It looks weaker only where the old test flattered it.